

Security+ Cloud Infrastructure Lab

*Building a Simulated AWS Environment Using Docker, LocalStack, Kali Linux,
Terraform, and AWS CLI*

Professional Student Lab Guide

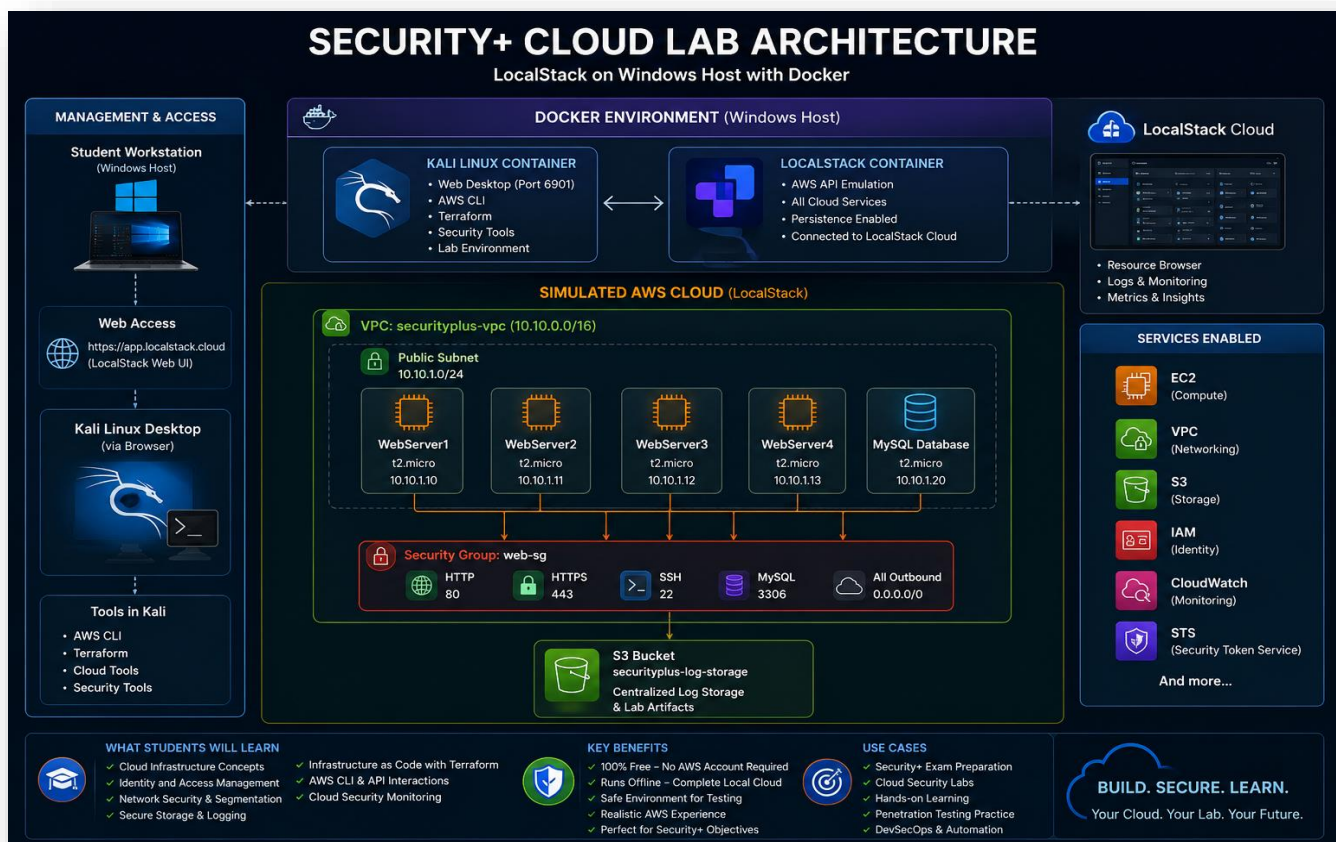
Lab Overview

In this lab students will build a simulated enterprise cloud environment that mimics core AWS services without requiring a real AWS account. Students will learn how cloud infrastructure is deployed, managed, secured, and automated using modern Infrastructure as Code techniques.

Learning Objectives

- Understand cloud infrastructure concepts
- Explain Infrastructure as Code (IaC)
- Deploy simulated AWS infrastructure
- Use Terraform to automate cloud resources
- Use AWS CLI to manage cloud infrastructure
- Create VPCs and subnets
- Deploy EC2 instances
- Create S3 cloud storage buckets
- Configure security groups
- Understand cloud security risks

Cloud Architecture



The following architecture will be built during this lab:

System Requirements

Requirement	Recommendation
Operating System	Windows 10 or Windows 11
RAM	16 GB Recommended
CPU	Quad-Core Processor
Disk Space	50 GB Free Space
Internet	Required for downloads

Part 1 — Install Docker Desktop

Docker Desktop provides the container platform used to run LocalStack and Kali Linux.

Step 1 — Download Docker Desktop

Navigate to the following website and download Docker Desktop for Windows:

<https://www.docker.com/products/docker-desktop/>

Step 2 — Install Docker Desktop

During installation ensure the following options are enabled:

- Use WSL2 backend
- Add Docker to PATH
- Enable virtualization support

Step 3 — Verify Docker Installation

Open Windows PowerShell and run:

```
docker --version
```

```
docker compose version
```

Part 2 — Create a Free LocalStack Account

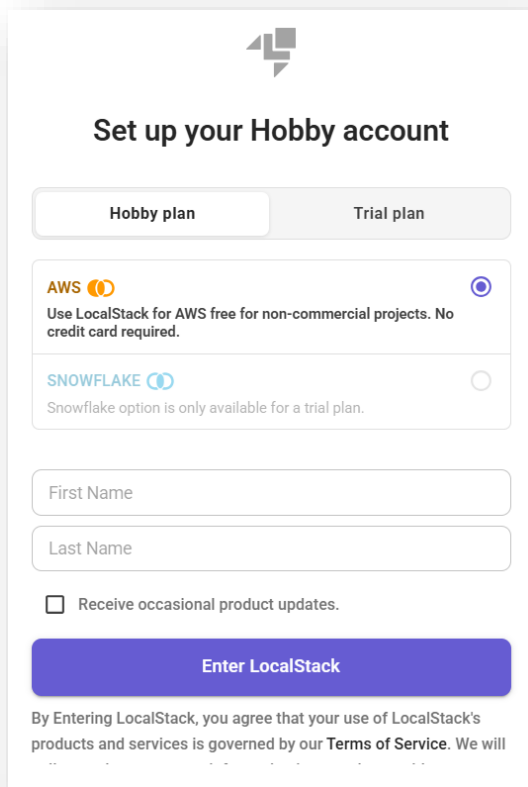
LocalStack simulates AWS cloud services locally on the Windows host system.

Step 1 — Create Account

Navigate to the following website:

<https://app.localstack.cloud/sign-up>

Create a free LocalStack account using your email address and select Hobby Plan



The screenshot shows the 'Set up your Hobby account' page. At the top is the LocalStack logo. Below it, the title 'Set up your Hobby account' is centered. There are two tabs: 'Hobby plan' (selected) and 'Trial plan'. Under the 'Hobby plan' tab, there are two options: 'AWS' (selected with a radio button) and 'SNOWFLAKE' (unselected). The 'AWS' option includes the text 'Use LocalStack for AWS free for non-commercial projects. No credit card required.' The 'SNOWFLAKE' option includes the text 'Snowflake option is only available for a trial plan.' Below these options are two input fields for 'First Name' and 'Last Name'. There is a checkbox labeled 'Receive occasional product updates.' and a large blue button labeled 'Enter LocalStack'. At the bottom, there is a disclaimer: 'By Entering LocalStack, you agree that your use of LocalStack's products and services is governed by our Terms of Service. We will'.

Set up your Hobby account

Hobby plan Trial plan

AWS Use LocalStack for AWS free for non-commercial projects. No credit card required.

SNOWFLAKE Snowflake option is only available for a trial plan.

First Name

Last Name

☐ Receive occasional product updates.

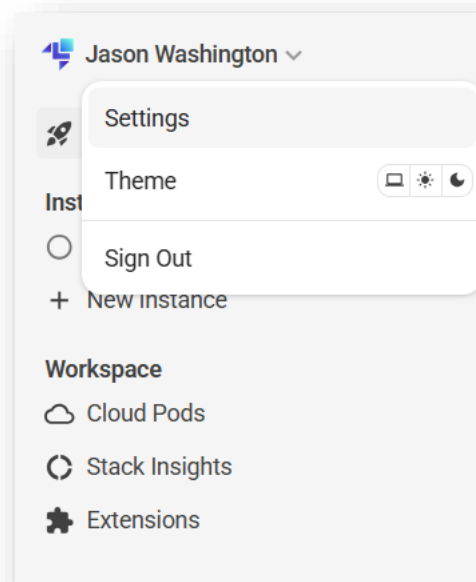
Enter LocalStack

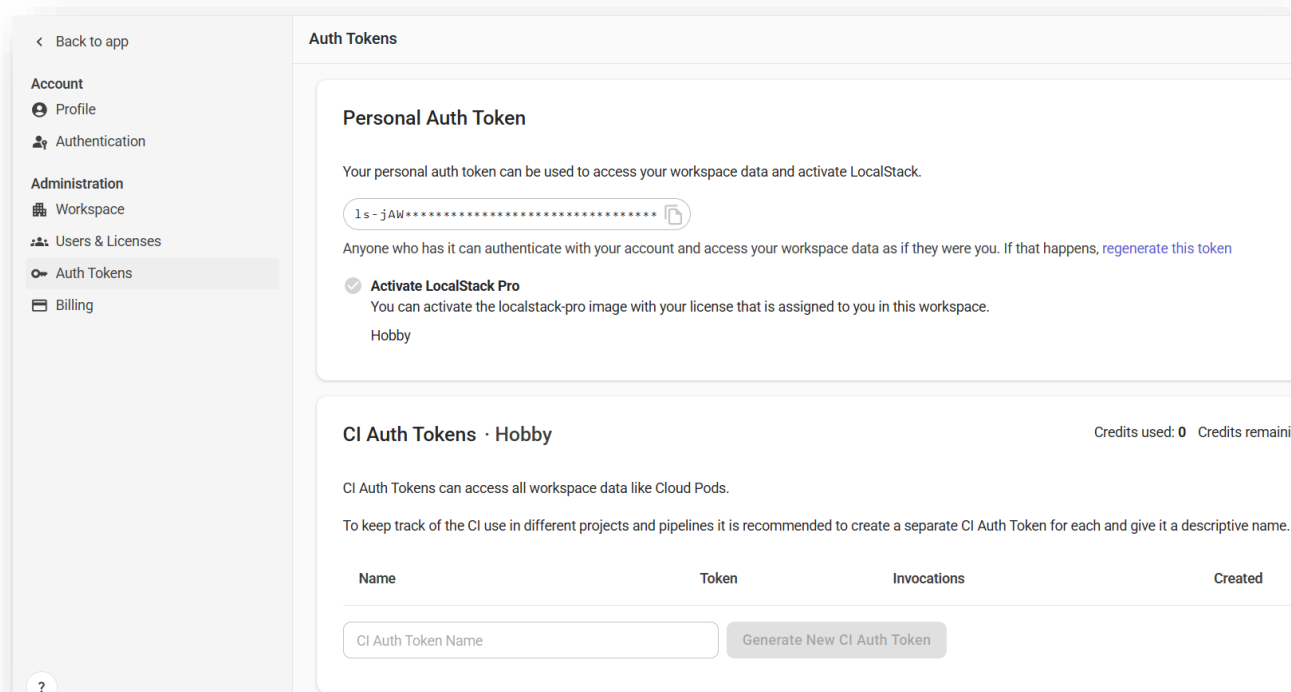
By Entering LocalStack, you agree that your use of LocalStack's products and services is governed by our [Terms of Service](#). We will

Step 2 — Generate Auth Token

After logging in:

- Click Profile
- Click Auth Tokens
- Click Create Token
- Copy the generated token





Part 3 — Create the Lab Folder Structure

Students will create the cloud lab folder structure on the Windows host PC.

Step 1 — Create Main Folder

```
mkdir C:\cloudlab
cd C:\cloudlab
```

Step 2 — Create Subfolders

```
mkdir terraform
mkdir localstack-data
mkdir docker-share
```

Part 4 — Create Docker Compose

Docker Compose will start both the LocalStack and Kali Linux containers.

Step 1 — Create docker-compose.yml

Inside C:\cloudlab create a file named docker-compose.yml.

Paste the provided instructor Docker Compose configuration into the file.

Part 5 — Start the Cloud Lab

Step 1 — Start Containers

```
docker compose up -d
```

Note: If the docker port need to be changed to something else. The change needs to be added to the docker-compose.yml as well as the terraform file.

Step 2 — Verify Running Containers

```
docker ps
```

Expected containers:

- localstack
- kali

You can check the localstack health by the following options:

curl http://localhost:7788/_localstack/health

or

Invoke-WebRequest http://localhost:7788/_localstack/health

Part 6 — Access Kali Linux Desktop

Step 1 — Open Kali Linux

```
https://localhost:6901
```

Use the following password:

```
password
```

Part 7 — Install AWS CLI and Terraform

The Kali Linux container will be used to manage cloud infrastructure.

Step 1 — Open Terminal

Inside Kali Linux open the terminal application.

Step 2 — Install AWS CLI

```
apt update
```

```
apt install -y awscli unzip curl wget jq
```

Step 3 — Install Terraform

```
wget https://releases.hashicorp.com/terraform/1.8.5/terraform_1.8.5_linux_amd64.zip
```

```
unzip terraform_1.8.5_linux_amd64.zip
```

```
mv terraform /usr/local/bin/
```

Part 8 — Configure AWS CLI

These are fake credentials used only for LocalStack.

```
aws configure
```

```
AWS Access Key ID: test
```

```
AWS Secret Access Key: test
```

```
Default region name: us-east-1
```

```
Default output format: json
```

Part 9 — Deploy Cloud Infrastructure

Terraform will create the following simulated cloud resources:

- 1 Virtual Private Cloud (VPC)
- 1 Public Subnet
- 1 Security Group
- 4 Web Servers
- 1 MySQL Database Server
- 1 S3 Log Storage Bucket

Step 1 — Create Terraform File

Inside the /terraform folder create a file named main.tf.

Paste the Terraform configuration provided by the instructor.

Step 2 — Initialize Terraform

```
cd /terraform
```

```
terraform init
```

Step 3 — Deploy Infrastructure

```
terraform apply
```

When prompted type:

```
yes
```

Part 10 — Verify Infrastructure

Students will verify that the infrastructure was successfully created.

List VPCs

```
aws --endpoint-url=http://localhost:4566 ec2 describe-vpcs
```

List EC2 Instances

```
aws --endpoint-url=http://localhost:4566 ec2 describe-instances
```

List S3 Buckets

```
aws --endpoint-url=http://localhost:4566 s3 ls
```

Part 11 — Security+ Concepts Demonstrated

- Cloud Infrastructure
- Shared Responsibility Model
- Infrastructure as Code
- Cloud APIs
- Virtual Networking
- Network Segmentation
- Security Groups
- Cloud Storage
- Identity and Access Management
- Cloud Automation

Instructor Discussion Topics

- Why public S3 buckets are dangerous
- Least privilege and IAM
- Why databases should not be public
- Flat networks vs segmented networks
- Infrastructure as Code advantages
- Cloud attack surface
- Automation and orchestration

Part 12 — Cleanup

Students should safely remove infrastructure after completing the lab.

terraform destroy

When prompted type:

yes

Conclusion

Students successfully deployed and managed a simulated enterprise cloud environment while learning core Security+ cloud infrastructure and cloud security concepts.

